

Technology Stack

Date	1 July 2026
Team ID	SWTID-2026-1614
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3 (Glassmorphism layout), Vanilla JavaScript (Jinja2 templates rendered by Flask)	The frontend utilizes Vanilla CSS to implement a premium glassmorphism form. Vanilla JavaScript handles the country search auto-fill logic and sends asynchronous Fetch API requests, avoiding the need for heavy frameworks.
2	Backend / Server-Side	Python, Flask	Flask is a lightweight WSGI web framework, well-suited for serving API prediction endpoints (such as POST /predict). Using Python ensures that backend routing and the ML models exist in the same environment.
3	Database / Data Storage	Flat-file CSV Database (HDI.csv) loaded in-memory	The application does not require transactional write operations. Pre-populated indicators for 195 countries are read directly from HDI.csv at runtime and cached, maintaining a stateless, memory-efficient backend.
4	Cloud / Hosting / Deployment	Render / PythonAnywhere (or Vercel Serverless with @vercel/python)	Standard, free-tier hosting for Python web apps. Automatically triggers rebuilds upon source repository updates without requiring manual server administration.
5	Version Control & CI/CD	GitHub	Used as the main repository host for team version tracking, enabling easy codebase backups, branch reviews, and integration with deployment hooks.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
6	Third-Party APIs / Other Tools	Scikit-Learn, Pandas, NumPy, Joblib	Scikit-Learn was used to train the baseline Linear Regression model, which is serialized using Joblib/Pickle. Pandas and NumPy prepare and format input matrices (Healthcare, Education, Income) for inference.